



Coding Thinker - MERN Stack Roadmap



MERN Stack Full Course

Build real-world skills across HTML, CSS, JavaScript, React, Node, Express, and MongoDB. This syllabus is structured, outcome-driven, and easy to track.

How to use this syllabus

- Skim the learning outcomes first
- Follow the study plan section by section
- Mark checkboxes to track progress
- Add links, code snippets, or screenshots under each day

Learning outcomes

- Write accessible, semantic HTML
- Build responsive, modern UIs with CSS, Flexbox, and Grid
- Program confidently in JavaScript, including async patterns
- Ship React apps with routing, state, and data fetching
- Create secure REST APIs with Node, Express, and MongoDB
- Integrate frontend and backend, then deploy to production

Prerequisites

- Basic computer literacy and Git installed
- Node.js LTS and a modern code editor (VS Code)

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Study plan

▼ HTML

- ☐ Day 1 · Introduction to Web & HTML
 - Web basics, client-server, HTML document structure
- ☐ Day 2 · Basic Tags
 - Headings, paragraphs, images, text formatting, inline vs block
- ☐ Day 3 · Hyperlinks & Lists
 - Anchor tags, ordered and unordered lists, nested lists, link states
- ☐ Day 4 · Tables
 - Structure, rowspan, colspan
- ☐ Day 5 · Forms I
 - Inputs, labels, textarea, select, fieldset, legend
- ☐ Day 6 · Forms II & Multimedia
 - Fieldset and legend, audio, video, iframe
- ☐ Day 7 · Semantic vs Non-Semantic HTML
 - header, nav, section, article, footer, div, span
- ☐ Day 8 · SEO Basics
 - Meta tags, favicon, accessibility basics

☐ Day 9 · Mini Project

- Personal info page
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▼ CSS

☐ Day 10 · Intro to CSS

- Syntax, selectors, inline vs internal vs external CSS

☐ Day 11 · Core Properties

- Colors, backgrounds, specificity

☐ Day 12 · Backgrounds

- Images and gradients: linear, radial, conic

☐ Day 13 · Text & Fonts

- Families, sizing, line height

☐ Day 14 · Box Model

- Padding, margin, border

☐ Day 15 · Styling Links, Lists, Tables

- Practical patterns

☐ Day 16 · Display

- inline, block, inline-block, none, visibility

☐ Day 17 · Position

- static, relative, absolute, fixed, sticky

☐ Day 18 · Float, Overflow, Z-index

- Layout control techniques

☐ Day 19 · Transitions & Transform

- duration, delay, timing, translate, rotate, scale, skew, matrix

☐ Day 20 · Flexbox I

- Container, direction, justify, align, wrap, gap, grow, shrink, basis

☐ Day 21 · Flexbox II

- Real-world patterns

- ☐ Day 22 · Animations
 - @keyframes, animation properties
 - ☐ Day 23 · CSS Grid I
 - Template rows/columns, gap, grid-row/column, area, place-items
 - ☐ Day 24 · CSS Grid II
 - Layout compositions
 - ☐ Day 25 · Media Queries
 - max-width, min-width, orientation, responsive design
 - ☐ Day 26 · Mini Project + Review
 - Portfolio layout
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▼ JavaScript

- ☐ Day 29 · JS Intro
 - Script loading, variables, console, alert, prompt
- ☐ Day 30 · Datatypes & Operators
 - Primitives, objects, arithmetic, comparison, logical, ternary
- ☐ Day 31 · Conditionals
 - if, else, switch
- ☐ Day 32 · Loops
 - for, while, do-while
- ☐ Day 33 · Functions
 - Declarations, expressions, params, returns, arrow functions
- ☐ Day 34 · Arrays & Methods
 - push, pop, shift, unshift, splice, slice, concat, forEach, map, filter, reduce
- ☐ Day 36 · Objects & Methods
 - keys, values, entries, assign, hasOwnProperty, freeze, seal
- ☐ Day 37 · String & Math
 - Common utilities, randoms

- ☐ Day 38 · Date & Time
 - Create and manipulate dates
- ☐ Day 39 · DOM Intro
 - Selecting and reading elements
- ☐ Day 40 · DOM Manipulation
 - Change content, styles, attributes
- ☐ Day 41 · Events
 - Event listeners and handlers
- ☐ Day 42 · Mini Project
 - Basic calculator
- ☐ Day 43 · Form Handling I
 - Validations with RegEx
- ☐ Day 44 · Form Handling II
 - Advanced patterns
- ☐ Day 45 · Execution & Hoisting
 - Phases, contexts, call stack, hoisting
- ☐ Day 46 · let/const & Scope
 - TDZ, block vs function vs global
- ☐ Day 47 · Lexical Environment & Closures
 - Scope chains, closure uses
- ☐ Day 48 · Higher-Order Functions
 - map, filter, reduce deep dive
- ☐ Day 49 · Timers & Callbacks
 - setTimeout, setInterval, callback patterns
- ☐ Day 50 · Callback Hell, Sync vs Async
 - Control-flow strategies
- ☐ Day 51 · Promises

- States, chaining
 - ☐ Day 52 · Promise API
 - all, race, allSettled, any
 - ☐ Day 53 · async/await & fetch
 - Async API calls
 - ☐ Day 54 · Error Handling & JSON
 - try...catch, JSON.parse/stringify
 - ☐ Day 55 · Mini Project
 - Fetch with async/await
 - ☐ Day 56 · Storage APIs
 - localStorage, sessionStorage CRUD
 - ☐ Day 57 · Review & Test
 - ☐ Day 58 · JS Major Project
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▼ Tailwind CSS

- ☐ Day 61 · Setup & Utilities
 - CDN/CLI, colors, spacing, typography
 - ☐ Day 62 · Layout Utilities
 - Flex, grid, container
 - ☐ Day 63 · Components & Responsive Design
 - Navbar, cards, buttons, shadows, hover, borders, breakpoints
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▼ React.js

- ☐ Day 64 · React Intro
 - Vite setup
- ☐ Day 65 · Components
 - Functional vs class
- ☐ Day 66 · JSX
 - Card component

- ☐ Day 67 · Styling
 - CSS, Tailwind, inline
- ☐ Day 68 · Props
- ☐ Day 69 · State (useState)
- ☐ Day 70 · Event Handling
 - onClick, onChange
- ☐ Day 71 · Mini Project
- ☐ Day 72 · Conditional Rendering
 - Ternary, short-circuit
- ☐ Day 73 · Lists & Keys
 - map in JSX
- ☐ Day 74 · Forms I
 - Controlled components
- ☐ Day 75 · Forms II
 - Validation and handling
- ☐ Day 76 · Lifting State Up
- ☐ Days 77–79 · Class Components
 - Lifecycle: mount, update, unmount
- ☐ Day 80 · useEffect
 - Side effects, API calls
- ☐ Day 81 · Fetch API in React
- ☐ Day 82 · Mini Project
- ☐ Days 83–84 · Routing
 - Basic and nested routes, Outlet, useNavigate
- ☐ Day 85 · Axios Mini Project
- ☐ Day 86 · Context API
 - Avoid prop drilling

- ☐ Day 87 · CRUD Project
 - Local storage or API
 - ☐ Day 90 · Redux Intro
 - Reducers, actions, store
 - ☐ Day 93 · React Major Project
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▼ Node.js

- ☐ Day 97 · Node Intro
 - Setup, npm
 - ☐ Day 98 · Modules
 - CommonJS vs ES Modules
 - ☐ Day 99 · Files & Paths
 - fs, path
 - ☐ Day 100 · http
 - Create a server
 - ☐ Day 101 · Event Loop
 - Async concepts
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▼ Express.js & MongoDB

- ☐ Day 102 · Express Intro
 - Server, routes
- ☐ Day 103 · Middleware
 - Built-in, custom
- ☐ Day 104 · REST API
 - GET, POST, PUT, DELETE
- ☐ Day 105 · Params & Query
 - URL parameters
- ☐ Day 106 · Postman
 - API testing

- ☐ Day 107 · Error Handling
 - Custom middleware
 - ☐ Day 108 · MongoDB Intro
 - Compass, CRUD basics
 - ☐ Day 109 · Mongoose Setup
 - Connect Node to MongoDB
 - ☐ Day 110 · Schema & Models
 - Define collections
 - ☐ Day 111 · MVC Pattern
 - Model, View, Controller
 - ☐ Day 112 · CRUD with Mongoose
 - ☐ Day 113 · Express + Mongo Integration
 - ☐ Day 114 · Validation
 - Mongoose validation
 - ☐ Day 115 · Authentication Intro
 - JWT, bcrypt
 - ☐ Day 116 · Signup & Login API
 - ☐ Day 117 · Authorization
 - Protect routes
 - ☐ Day 118 · Environment Variables
 - dotenv
 - ☐ Day 119 · File Uploads
 - Multer
 - ☐ Day 120 · Mini Backend Project
 - Blog or Bookstore API
 - ☐ Day 122 · Testing & Debugging
 - Postman and error fixes
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▼ MERN Integration & Deployment

- ☐ Day 124 · Project Intro
 - Folder structure
 - ☐ Day 125 · Frontend ↔ Backend Connect
 - Axios + proxy
 - ☐ Day 126 · CRUD Integration
 - ☐ Day 127 · React Authentication
 - JWT + Context
 - ☐ Day 128 · Protected Routes
 - ☐ Day 129 · Profile & Dashboard
 - ☐ Day 130 · Project Styling
 - Tailwind UI
 - ☐ Day 131 · Testing & Debugging
 - End-to-end checks
 - ☐ Day 132 · Deployment
 - Production steps
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Tips for success

- Keep projects small and frequent
- Write down stumbling blocks and revisit weekly
- Prioritize readability over cleverness
- Practice with real APIs and sample data
- Refactor and document your code

Schedule and timeline

- Total sessions in this syllabus: **124**
- Class length: **1 hr 20 min** per session
- Total time required: **165 hrs 20 min**

- Weekly pace: **3 classes/week** → **4 hrs/week**
- Estimated duration to complete: **≈ 41.3 weeks**
 - That's 41 weeks + 1 extra class
- Calendar estimate: **≈ 9.6 months** at this pace
 - Rounded guidance: plan for **9.5–10 months**

Notes

- Includes lessons, practice, and mini/major projects at the listed pace
- Add a 10–15% buffer for revision, exams, or breaks → **~10–11 months** total